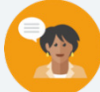


7.7 Factoring Perfect Square Trinomials

Lesson Introduction

 Benchmark	MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.		 Learning Target	I can factor perfect square trinomials.
 Benchmark Coherence	Building On MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.		Working Toward MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system.	
 Essential Questions	- How can you determine if a trinomial is a perfect square trinomial? - How can recognizing that a polynomial is a perfect square trinomial help with factoring?			
 Lesson Narrative	In this lesson, students will be introduced to the concept of a perfect square trinomial, learning what they are and how to identify them. They will then transition into learning how to factor a perfect square trinomial before practicing identifying and factoring perfect square trinomials.			
 Component Summary	7.7.1 Warm-Up (~5 min.)	Create equivalent forms of expressions (<i>SE p. 391</i>)	7.7.4 Guided Practice (10 min.)	Guided practice with factoring perfect square trinomials (<i>SE p. 396</i>)
	7.7.2 Collaboration (10-15 min.)	Use algebra tiles to explore perfect square trinomials (<i>SE p. 392</i>)	7.7.5 Your Turn! (5 min.)	Independent practice with factoring perfect square trinomials (<i>SE p. 396</i>)
	7.7.3 Guided Instruction (10 min.)	Formal instruction on factoring perfect square trinomials (<i>SE p. 395</i>)	7.7.6 Wrap-Up (~5 min.)	Emoji self-reflection activity (<i>SE p. 397</i>)
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> Perfect square trinomial <u>Additional Terms:</u> N/A		For 7.7.4 Guided Practice, consider creating an index card with the answers or a printout of the table to quickly identify where students are making errors during the practice while circulating the room.	

 Teacher Tips	Common Misconceptions - Students may believe any larger numbers or coefficients cannot be perfect square trinomials (like question 1 in Your Turn!). - Students may mistake a difference of squares for a perfect square trinomial.	Things to Consider Students should not see perfect square trinomials as something else or new or an addition to what they've learned. The purpose is for students to discover the relationship between a perfect square trinomial and its factored form. Students can still use other methods for factoring these expressions.
	Literacy and Language Routines Compare and Connect 7.7.3 Guided Instruction provides students with guided instruction on factoring perfect square trinomials. During this activity, ensure that students can hear your thinking aloud during instruction. This way, students can connect this thinking to the content from start to finish.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection 7.7.2 Collaboration provides students with an activity where they work with a partner to explore perfect square trinomials using algebra tiles. Students will communicate mathematical ideas, vocabulary, and methods effectively. Consider ways to establish a culture in which students ask questions of the teacher and their peers and error is seen as an opportunity for learning.
	 Differentiate Instruction For 7.7.2 Collaboration, students will work together. Consider grouping students based on their learning styles. This can help with the way students connect to the concept of perfect square trinomials (i.e., with physical algebra tiles). Consider providing physical algebra tiles or a paper cut-out version in order to provide a physical manipulative for the algebra tile portions of the activity. 7.7.5 Your Turn! provides students with an independent practice opportunity with factoring perfect square trinomials. Ensure that students show their work so you can see if they recognize that they are perfect square trinomials or if they are factoring as a standard trinomial.	
Lesson Notes:		

7.7.1 Warm-Up: Critical Thinking

Component Overview: Students will complete a table with equivalent quadratic expressions in standard and factored form.



7.7.1 Warm-Up: Critical Thinking

1. Fill in the blanks with an integer so that the quadratic expression is factorable. Then, complete the factored form.

Answers will vary.

Quadratic Expression	Factored Form
$x^2 + 12x + 36$	$(x + 6)(x + 6)$
$x^2 - 9x - 36$	$(x - 12)(x + 3)$
$x^2 + 9x - 36$	$(x - 3)(x + 12)$



Use the first expression as a launch into perfect square trinomials.

- How is the first expression different than the rest?
- Is there a way to efficiently identify when that may happen?
- What does that make you think about factoring?

7.7.2 Collaboration: Perfect Square Trinomials

Component Overview: Students will work collaboratively to discover the concept of perfect square trinomials. This discovery will be done using algebra tiles. For this activity, students should work in pairs or small groups to complete the activity.



7.7.2 Collaboration: Perfect Square Trinomials

- The quadratic expressions represented by algebra tiles are perfect square trinomials.

A	B

- Discuss with your partner why these quadratic expressions are called perfect square trinomials.
Answers will vary. The tiles are laid out in a square.
- Write the trinomials represented by the algebra tiles.

Trinomial A	Trinomial B
$x^2 + 6x + 9$	$x^2 - 8x + 16$

- Use the algebra tile representations to write the factored form of each trinomial.

Factored Form of Trinomial A	Factored Form of Trinomial B
$(x + 3)(x + 3)$	$(x - 4)(x - 4)$



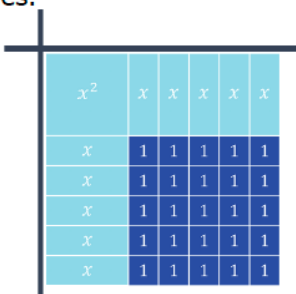
For this activity, release students to work with a partner or small group to complete each part. For part a, this is a discussion question with less focus on writing their entire description in the book. When reviewing the answers, have multiple partner pairs or groups share their responses and rationales.



How did you determine the factored form of Trinomials A and B?

7.7.2 Collaboration: Perfect Square Trinomials (continued)

2. The quadratic expression $x^2 + 10x + 25$ is represented by the algebra tiles.



- a. Pranjali used the algebra tiles to write the factored form of $x^2 + 10x + 25$ as $(x + 5)^2$.

Check Pranjali's work by completing the steps.

$$(x + 5)^2 = (x + 5)(x + 5)$$

$$(x + 5)(x + 5) = (x + 5)^2$$

- b. Discuss with your partner how writing the factored form of the quadratic expression $x^2 + 10x + 25$ as $(x + 5)^2$ relates to the expression being called a perfect square trinomial. Summarize your discussion.

Answers will vary. The binomial is squared.



If available, consider providing students with physical or virtual algebra tiles. This can help learners have a hands-on entry point into this question and others to explore the information provided in this activity.



Agree or Disagree

Highlight student responses that showcase conceptual understanding by pausing for a poll to agree or disagree, focusing on the rationale behind your decision rather than the decision itself.

7.7.2 Collaboration: Perfect Square Trinomials (continued)

3. Work with your partner to determine which quadratic expressions are perfect square trinomials.

Expression	Algebra Tiles
$4x^2 - 9$ Is it a perfect square trinomial? ☐ Yes ☒ No	
$4x^2 + 12x + 9$ Is it a perfect square trinomial? ☒ Yes ☐ No	
$4x^2 - 12x + 9$ Is it a perfect square trinomial? ☒ Yes ☐ No	
$9x^2 + 6x + 1$ Is it a perfect square trinomial? ☒ Yes ☐ No	

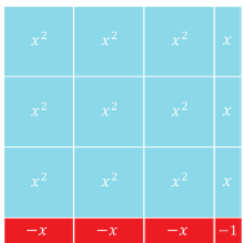


What about this expression makes it not a perfect square trinomial?



Highlight the difference between the difference of squares (no need to use this vocabulary yet, as this will be introduced in Lesson 8) and perfect square trinomials. Some students may be confused on the first one because there are no obvious “holes” in the mat. It appears to be a square even though its length and width are not the same.

7.7.2 Collaboration: Perfect Square Trinomials (continued)

$9x^2 - 1$ Is it a perfect square trinomial? ☐ Yes ☒ No	
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4. Describe some key characteristics of perfect square trinomials by relating both the representation of a quadratic expression using algebra tiles and the trinomial.

Answers will vary. A perfect square trinomial will create a square when representing it using algebra tiles. The a -term and the c -term are perfect squares.



How can you tell, by looking at the algebra tiles, that this is not a perfect square trinomial?



When reviewing answers, use clear and concise language to highlight how the algebra tiles show that a perfect square trinomial exists.

7.7.3 Guided Instruction: Factoring Perfect Square Trinomials

Component Overview: Students will receive formal instruction on factoring perfect square trinomials. They will be introduced to the definition of a perfect square trinomial and the process for factoring them.



7.7.3 Guided Instruction: Factoring Perfect Square Trinomials

A **perfect square trinomial** is a trinomial of the form $a^2 + 2ab + b^2$ or $a^2 - 2ab + b^2$.

1. The quadratic expression $4x^2 + 12x + 9$ is a perfect square trinomial. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.
 $(2x)^2 + 2(2x)(3) + 3^2$

Perfect square trinomials of the form $a^2 + 2ab + b^2$ can be factored as $(a + b)^2$.

2. Factor $4x^2 + 12x + 9$.
 $(2x + 3)^2$

3. The quadratic expression $9x^2 - 12x + 4$ is a perfect square trinomial. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.
 $(3x)^2 - 2(3x)(2) + 2^2$

Perfect square trinomials of the form $a^2 - 2ab + b^2$ can be factored as $(a - b)^2$.

4. Factor $9x^2 - 12x + 4$.
 $(3x - 2)^2$



Add the definition of a perfect square trinomial to any existing anchor charts, Frayer Models, or guided notes that students have for the factoring processes learned in this unit.



Students have intentionally already explored perfect square trinomials. They now have prior knowledge to activate during this activity rather than approaching instruction as something completely new. Use inquiry to activate connections from the Collaboration activity.

- $4x^2$ is a perfect square. What is the square root of $4x^2$? This is the value of a .
- Which other term in the given expression is a perfect square? What is its square root? This is the value of b .
- What is a way to determine if a trinomial would be classified as a perfect square trinomial?
- What is the benefit to the factoring process by knowing that a trinomial is a perfect square trinomial?

7.7.4 Guided Practice: Factoring Perfect Square Trinomials

Component Overview: Students will practice factoring perfect squares by identifying, given a quadratic expression, whether it meets the conditions of a perfect square trinomial.



7.7.4 Guided Practice: Factoring Perfect Square Trinomials

- Place a check mark in the appropriate columns of the table to indicate whether or not each quadratic expression is a perfect square trinomial. If it is a perfect square, write its factored form as $(a + b)^2$ or $(a - b)^2$.

Quadratic Expression	First Term is a Perfect Square	Third Term is a Perfect Square	Second Term is $2ab$ or $-2ab$	Factored Form
$2g^2 + 6g + 9$		✓		<i>Prime</i>
$16y^2 - 56y + 49$	✓	✓	✓	$(4y - 7)^2$
$b^2 - 26b + 25$	✓	✓		$(b - 25)(b - 1)$
$4w^2 - 16w + 8$	✓			$4(w^2 - 4w + 2)$
$9x^2 - 4$	✓	✓		$(3x + 2)(3x - 2)$
$q^2 - 2qz + z^2$	✓	✓	✓	$(q - z)^2$



Consider creating a small version of the table answer key or printing a small version to keep handy. This will be helpful when quickly scanning student work to see where potential errors are being made to intervene more quickly.

7.7.5 Your Turn!

Component Overview: Students will work independently to factor two perfect square trinomials.



7.7.5 Your Turn!

1. Factor $16h^2 + 88h + 121$.
 $(4h + 11)^2$
2. Factor $k^2 - 10k + 25$.
 $(k - 5)^2$



For this activity, allow students to try these on their own to determine their level of mastery of how to factor these types of trinomials.

7.7.6 Wrap-Up: Reflection

Component Overview: Students will complete an emoji-scale self-reflection activity on their understanding of perfect square trinomials.



7.7.6 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of factoring perfect square trinomials.

Answers will vary.



0 1 2 3 4 5 6 7 8 9 10

2. Complete the statement.

Answers will vary.

I chose this emoji because _____

_____.



Prompt students to pair their responses with an example that demonstrates their level of understanding. Students who mark 1-5 on the scale should use Mark-Up text to identify at which point they have questions, while students who mark 6-10 can show an example or find a partner who marked 1-5 and assist.